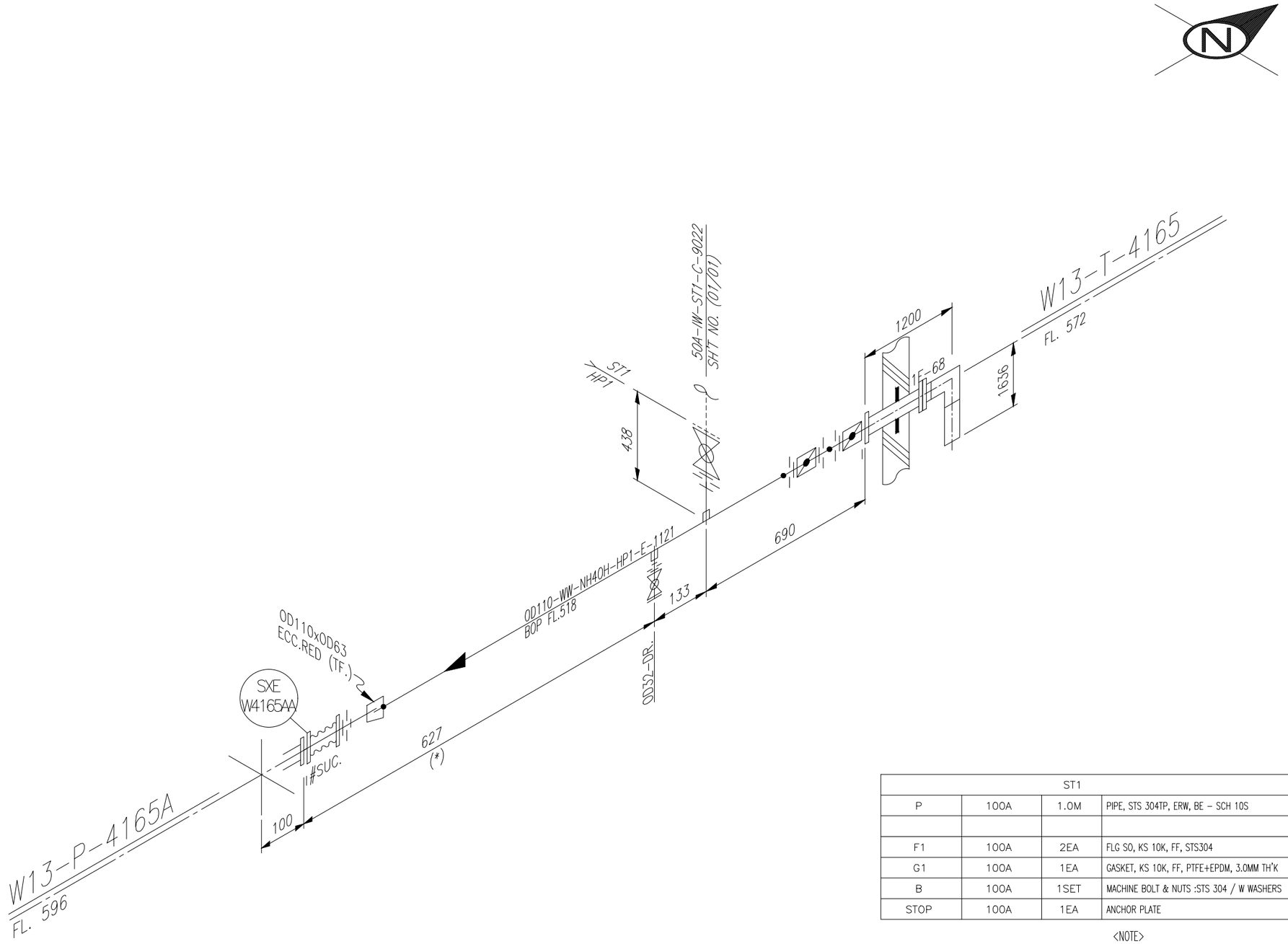


ST1			
P	100A	1.0M	PIPE, STS 304TP, ERW, BE – SCH 10S
F1	100A	2EA	FLG SO, KS 10K, FF, STS304
G1	100A	1EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	100A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
STOP	100A	1EA	ANCHOR PLATE

<NOTE>
1. (*) FIELD ADJUST

AS BUILT

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		HP1	
P	OD110	1.1M	PIPE, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
P	OD63	0.3M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD32	0.1M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
RE	OD110xOD63	1EA	REDUCER ECC, HDPE PE100, PE, BUTT FUSION,
			PN16/SDR11
SD	OD110xOD63	1EA	SADDLE, HDPE PE100, PE, BRANCH-SOCKET FUSION,
			PN16/SDR11
SD	OD110xOD32	1EA	SADDLE, HDPE PE100, PE, BRANCH-SOCKET FUSION,
			PN16/SDR11
VB	OD63	1EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP /
			TRIM-PP / SEAT-PTFE, FB, LEVER
VB	OD32	1EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP /
			TRIM-PP / SEAT-PTFE, FB, LEVER
VF	OD110	2EA	BUTTERFLY VALVE, KS 10K, WAFER FF, BODY-PP /
			TRIM-PP / EPDM-LINED, GEAR
F	OD110	3EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
			LAP JOINT-STS304 / STUB END-HDPE PE100
F	OD63	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
			LAP JOINT-STS304 / STUB END-HDPE PE100
F	OD32	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
			LAP JOINT-STS304 / STUB END-HDPE PE100
FB	OD32	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STS304
			/ LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD110	4EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD63	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD32	1EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
BL	OD110	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)
B	OD63	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	OD32	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
FLXR	OD63	1EA	FLEXIBLE JOINT(TPC RUBBER TYPE)
IN	25txOD110	1.1M	관 보온재(고무발포)
IN	25txOD75	0.1M	관 보온재(고무발포)
IN	25txOD63	1.1M	관 보온재(고무발포)
IN	25txOD32	0.1M	관 보온재(고무발포)
		FR1	
P2	100A	1.3M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.0T
E90A	100A	1EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.0T
F2	100A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	0.4	60
OPERATING	ATM	AMB

AREA	W13
UNIT	1FA
REF. PLAN DWG.	CJ-M15W13-PI-A30-010001-A
REF. P & ID	CJ-M15W13-PR-A20-000031

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
	26.04.30	AS-BUILT	황래종	안남후	박동영

CLIENT
SK hynix

CONTRACTOR
SK ecoplant

PROJECT NAME
M15X Ph3 Project

COMMODITY
WW-NH4OH

JOB NO.
22330D

LINE NO.
OD110-WW-NH4OH-HP1-E-1121

SH'T NO.
01 / 01

REV.

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